Dr. Yilmaz

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Project “Solar Reflector”

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Pd. 7 Pd. 3

Background

- Heliostat: A device that uses mirrors & motors to track the sun's movement across sky & reflect it.

[4i]

The Problem

- Good news: Sunlight reaches most locations of the earth within 24 hours.
- Bad news: Shadows leave areas of the planet in the dark and move throughout the day.

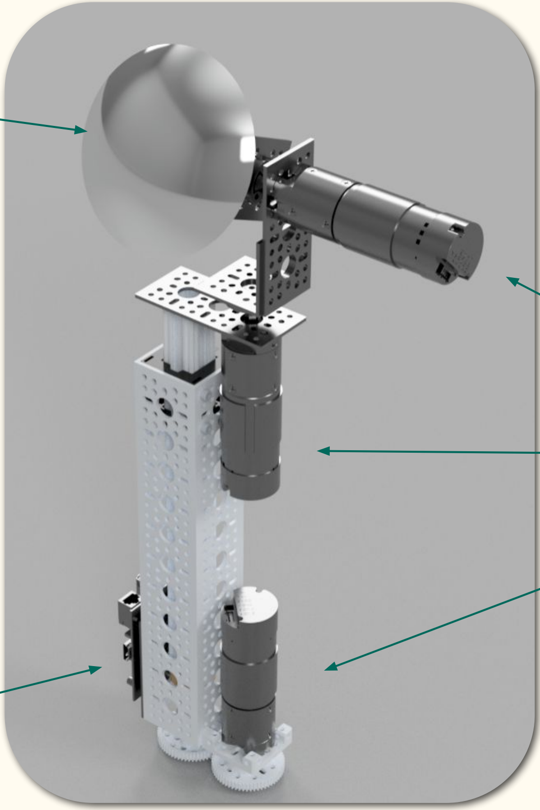
[2i]

[1i]

When the sun is on the other side of the roof, the panels receive less light. Even during the day!

Our Solution: The Micro-Heliostat

Convex Mirror

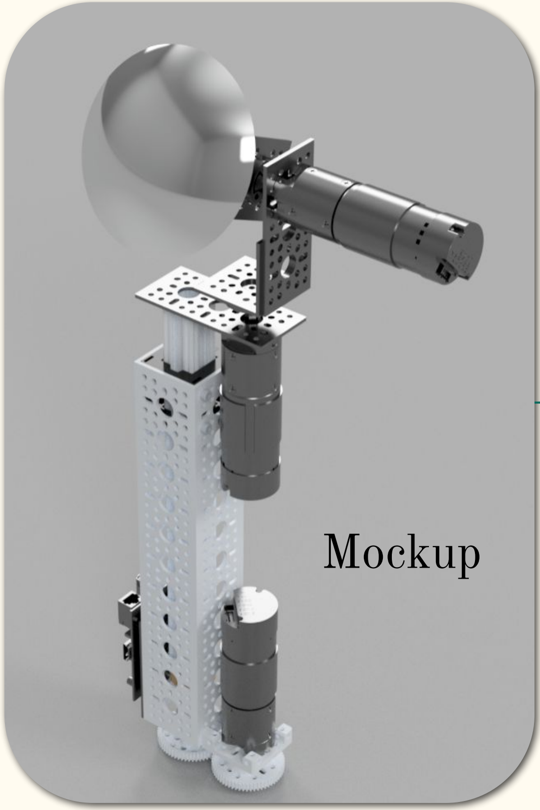


This image is our proof-of-concept from October

Positioning Motors

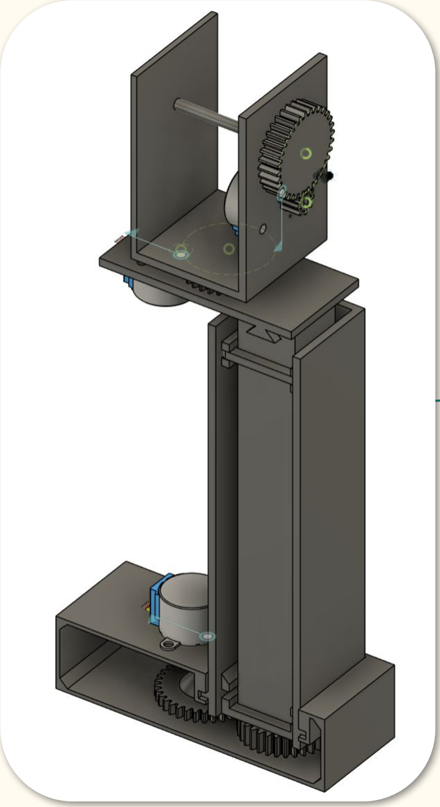
Raspberry Pi

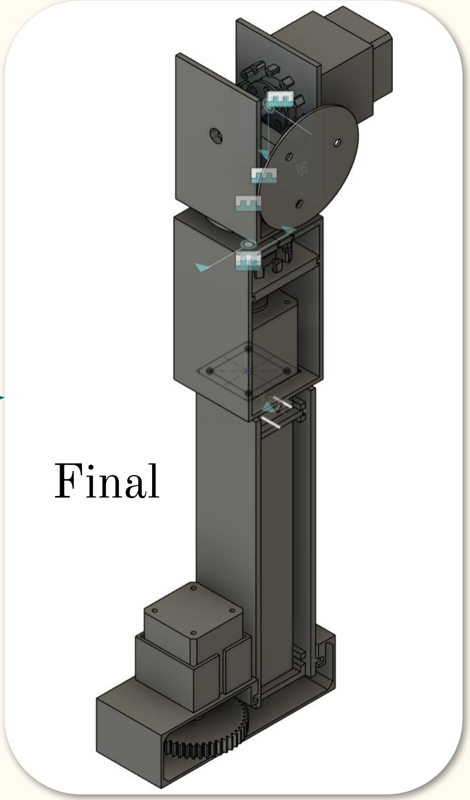
The Process



Mockup

First





Final

OTHER SOLUTIONS

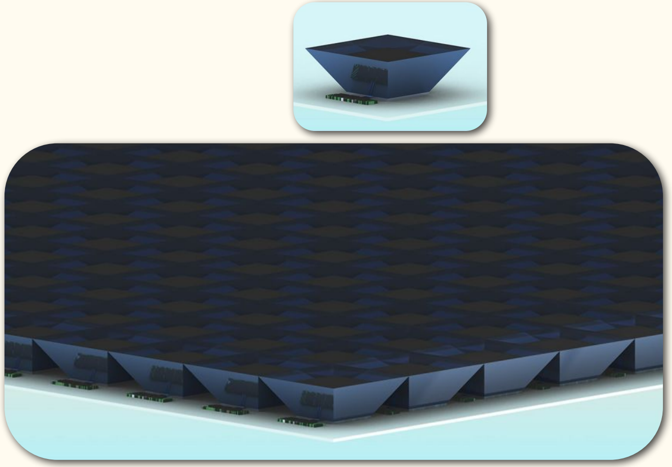
Lighting specific:

- Light Bulbs/LEDs
- Stationary & flat mirrors [1]
- Large Heliostats [3]
- Wikoda Sunflower (missing height movement)

Solar Panel specific:

- Moving solar panel to face the sun
- Axially Graded Index Lens (Stanford) [2]

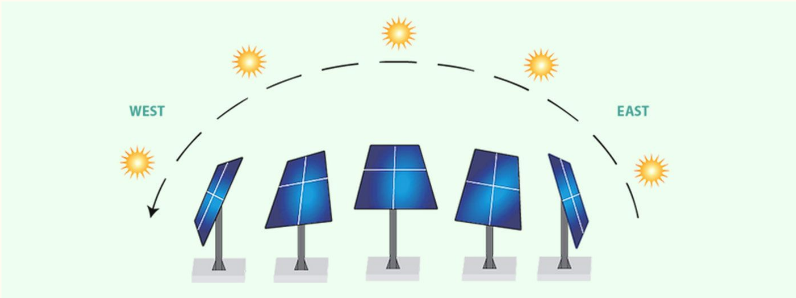
Wikoda Sunflower



Stanford's AGILE array system [2]

Novelty

- Tracks sun direction (unlike stationary mirrors) for dynamic alignment
- Compact and lightweight heliostat design
- Three degrees of freedom: pitch, yaw, and height adjustment
- Modular mechanical and electronic system architecture
- Wide range of potential applications across energy and lighting systems



Theoretical Impact

- Increase solar panel efficiency by redirecting sunlight throughout the day
- Reduce daytime indoor lighting usage through natural light reflection
- Improve sunlight access in shaded residential or urban environments
- Support sustainable and energy-efficient solar tracking solutions

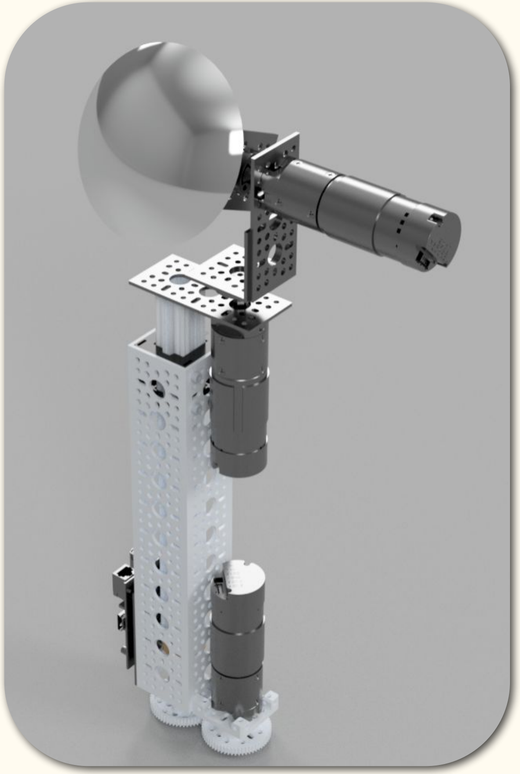
Limitations

- User must frequently clean the mirror manually
- Reflector requires an elevated location & sunny conditions
- Will require a CONSISTENT power source, so using solar power will not be sufficient

METHOD: Mechanics

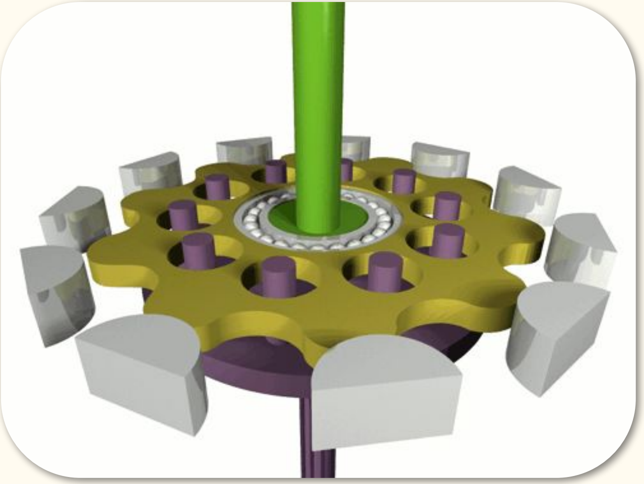
Parts and purposes:

- Stepper Motor x3: Pitch, Yaw, and Lift
- Convex Mirror: Reflection
- Raspberry Pi 4: running code & grabbing sun data online
- Cycloidal drive system: 3D printed gear setup that creates ten-fold increases in torque.
- Linear Actuator: Allows for high-torque height adjustment.



The Torque God

- In between our first iteration and our final, we made some great changes because of one enemy. Torque.
- We switched out the smaller stepper motors for NEMA-17 steppers, in order to fight against the torque caused by gravity.
- NEMA-17s alone could not solve our problems, so we used cycloidal drives.
- They provide an extremely efficient way to get high torque ratios. A space in which you could get a 4 times torque increase with normal gears could give a 10 times torque increase as seen to the right.

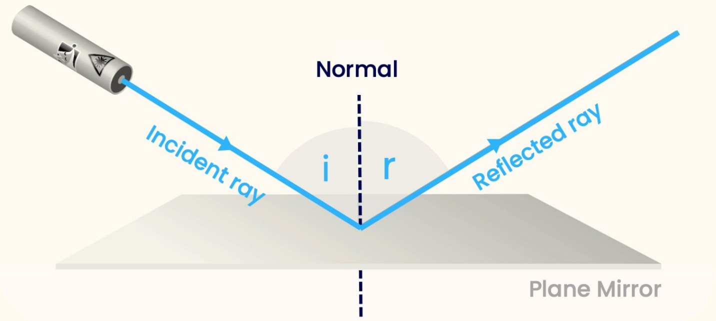


METHOD: Programming

Program:

- UI allows user to set desired position & current location
- Code "tares" position of steppers upon startup
- Uses the IPGeolocation API to obtain real-time sun data based on user location
- Pi controls the motors via GPIO pins
- Pi rotates the mirror in order to aim sunlight at a predetermined region using a function
- Pi continuously runs whether UI program is running or not

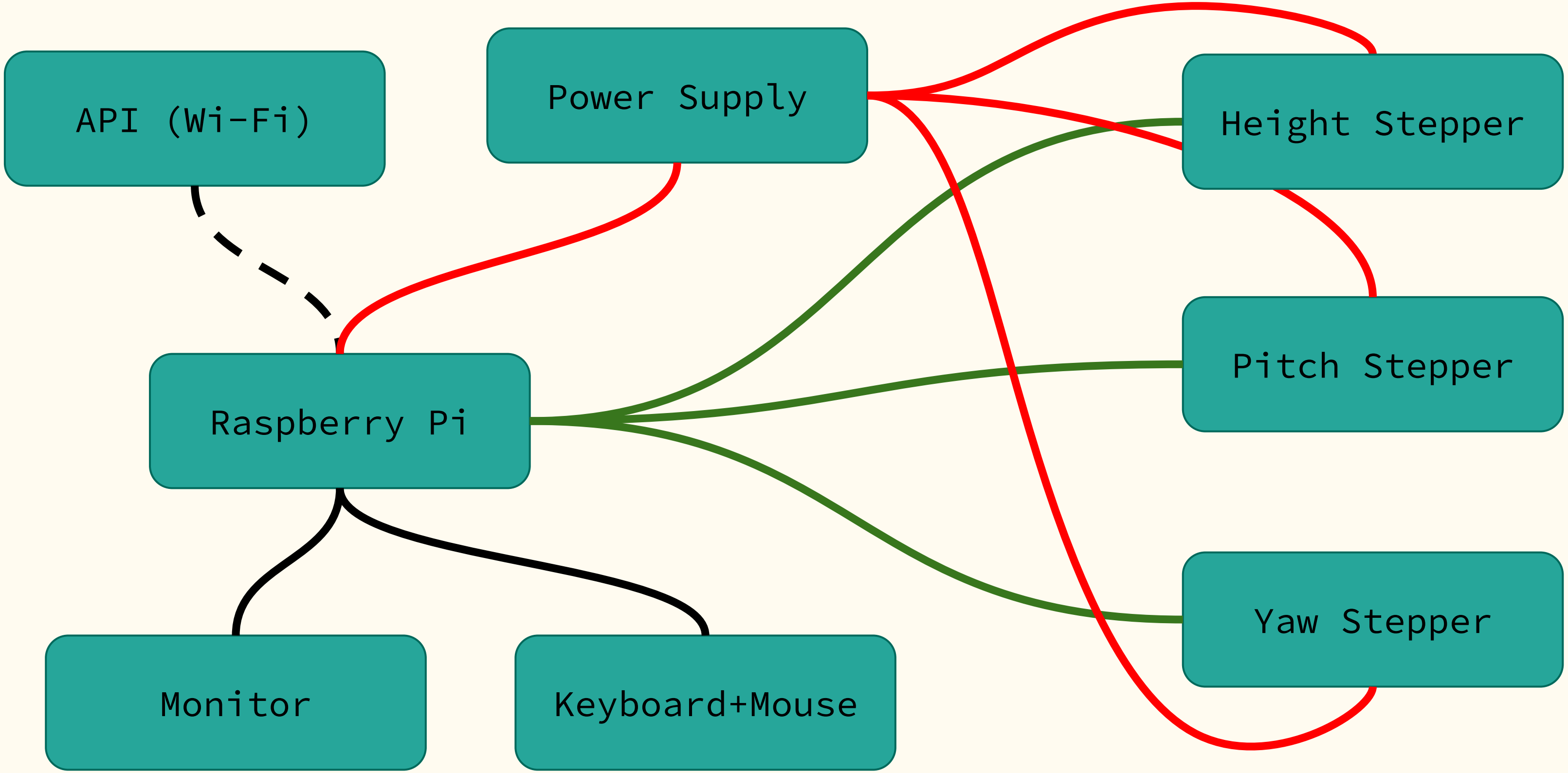
[5i]



i=r

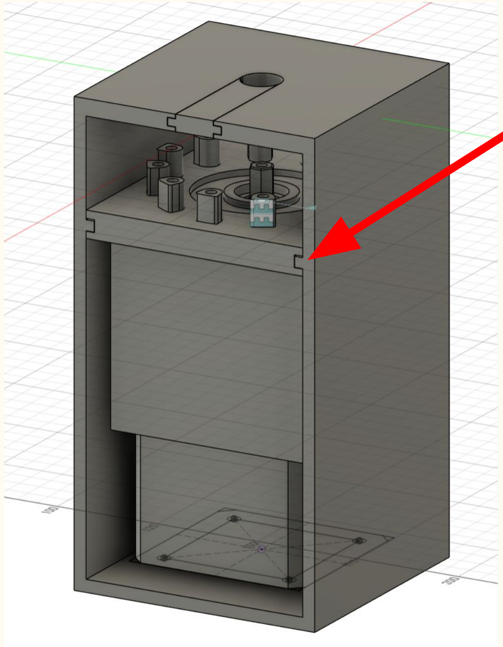
We applied this concept to 3D

System Architecture Diagram



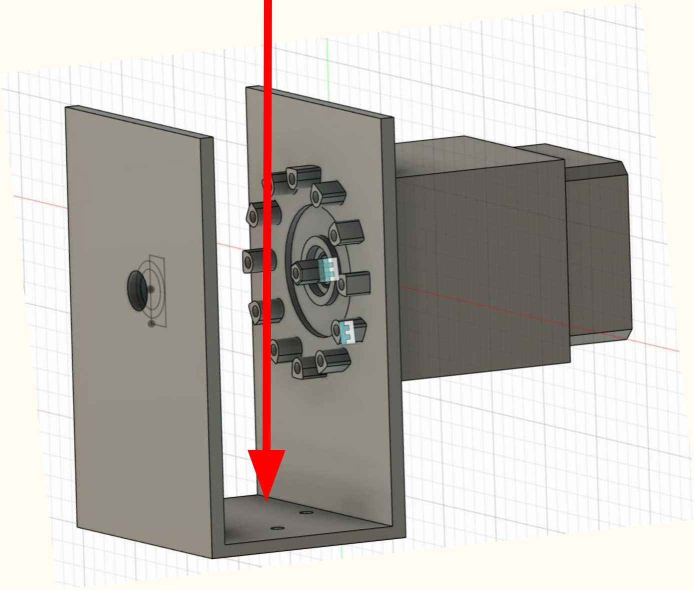
Designs

Modular design makes motor swaps and repairs simple



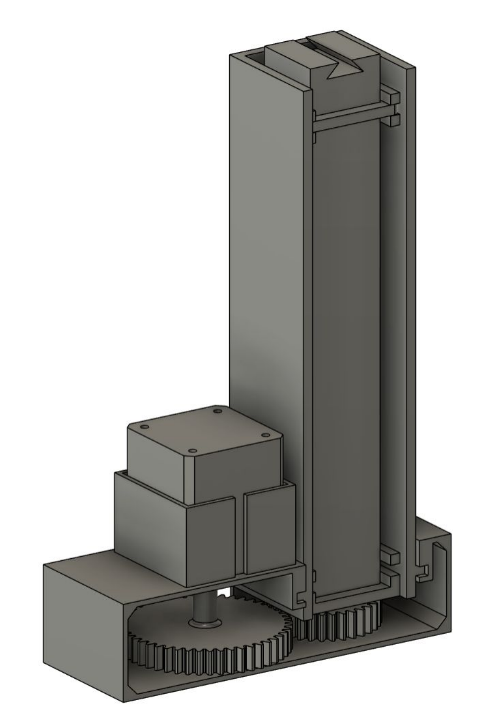
Yaw Control System

Both use a 1:11 gear ratio

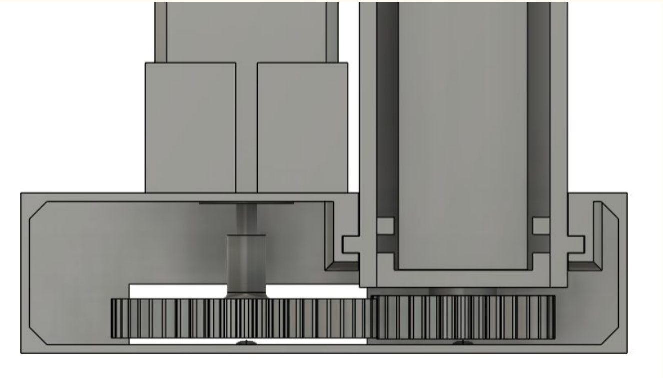


Pitch System

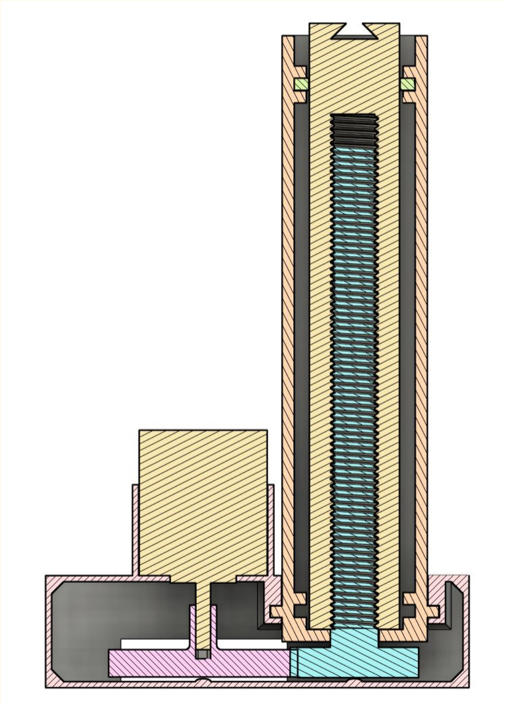
Prototype designs



Linear Actuator

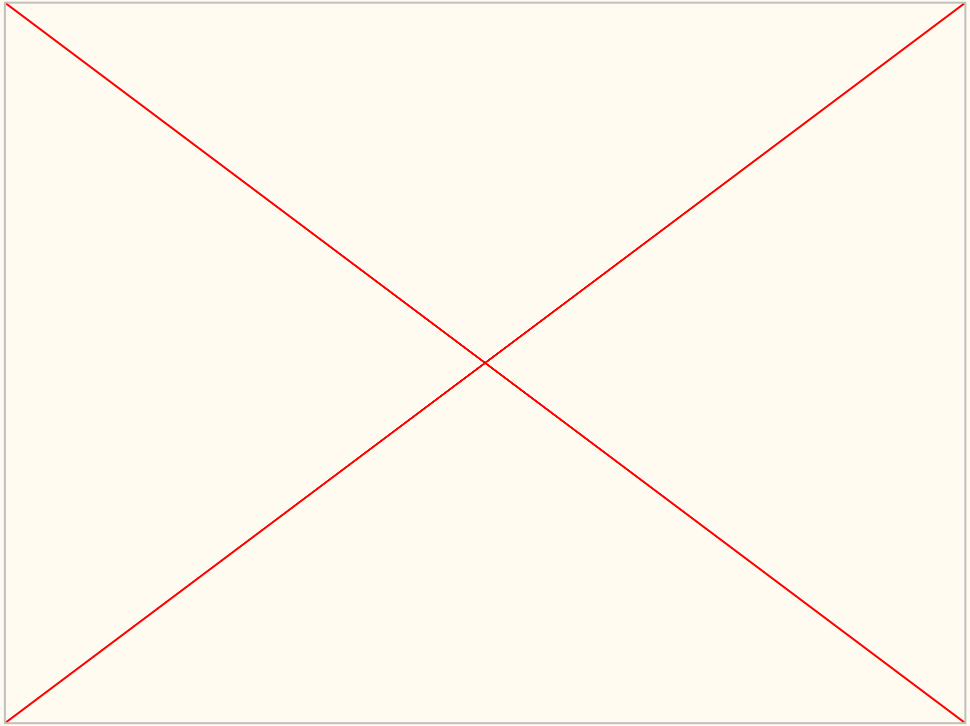


Gear View

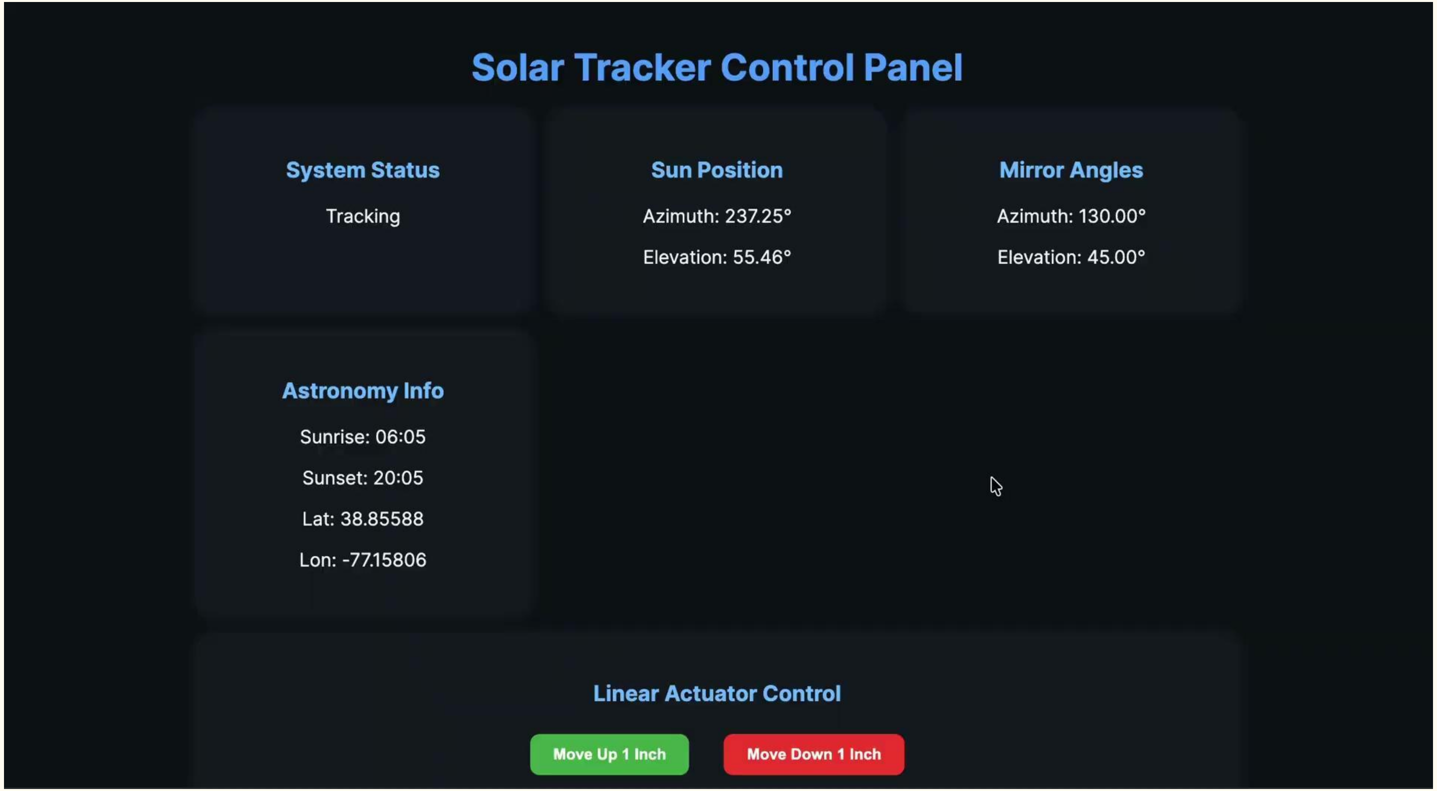


Cross-Section

Demo



UI DEMO


Solar Tracker Control Panel
System Status
Tracking
Sun Position
Azimuth: 237.25°
Elevation: 55.46°
Mirror Angles
Azimuth: 130.00°
Elevation: 45.00°
Astronomy Info
Sunrise: 06:05
Sunset: 20:05
Lat: 38.85588
Lon: -77.15806
Linear Actuator Control
Move Up 1 Inch
Move Down 1 Inch

CONCLUSION AND FUTURE WORK

Conclusion:

- We designed and built a micro-heliostat with extra functionalities so that it may serve multiple purposes.

Future work:

- Switch from plastics to metals to improve frame and structure
- Wire management/cleanup to keep connections intact
    - Preferably sautered connections instead of using breadboards
- Iris system: covers mirror at night, reduces dust
- User placement recommendations in UI

[3i]

REFERENCES

[1] https://www.researchgate.net/publication/263967079_Increase_the_Efficiency_of_Solar_Panel_by_using_Mirrors

[2] https://news.stanford.edu/stories/2022/06/new-optical-device-help-solar-arrays-focus-light-even-clouds

[3] https://www.sciencedirect.com/science/article/pii/S0038092X21001122

Research (Used for Helping us decide on this project idea)

https://www.sciencedirect.com/science/article/pii/S0360132316303766

https://www.sciencedirect.com/science/article/pii/S0038092X25001021

https://www.sciencedirect.com/science/article/pii/S0960148125002745

Images

[1i] https://www.pexels.com/photo/solar-panels-on-house-roof-17762230/

[2i] https://www.rawpixel.com/image/6481112/image-public-domain-sun-illustrations

[3i] https://www.tanerxun.com/product/3dpd50b5ty4-b-3d-printable-rotating-shutter-mechanical-iris-aperture?srsltid=AfmBOopikmQtD58oxel9AxcK0dDyGceDVqN3eqKq4lD8Vxa9db9CBLDe

[4i] https://commons.wikimedia.org/wiki/File:CSIRO_ScienceImage_3779_Heliostat_mirror_field_at_the_National_Solar_Energy_Centre_in_Newcastle_NSW.jpg

[5i] https://www.physicsfox.org/waves/reflection/

[6i] https://intello.co.in/understanding-solar-tracking-systems-its-working-types-pros-and-cons/

Q&A